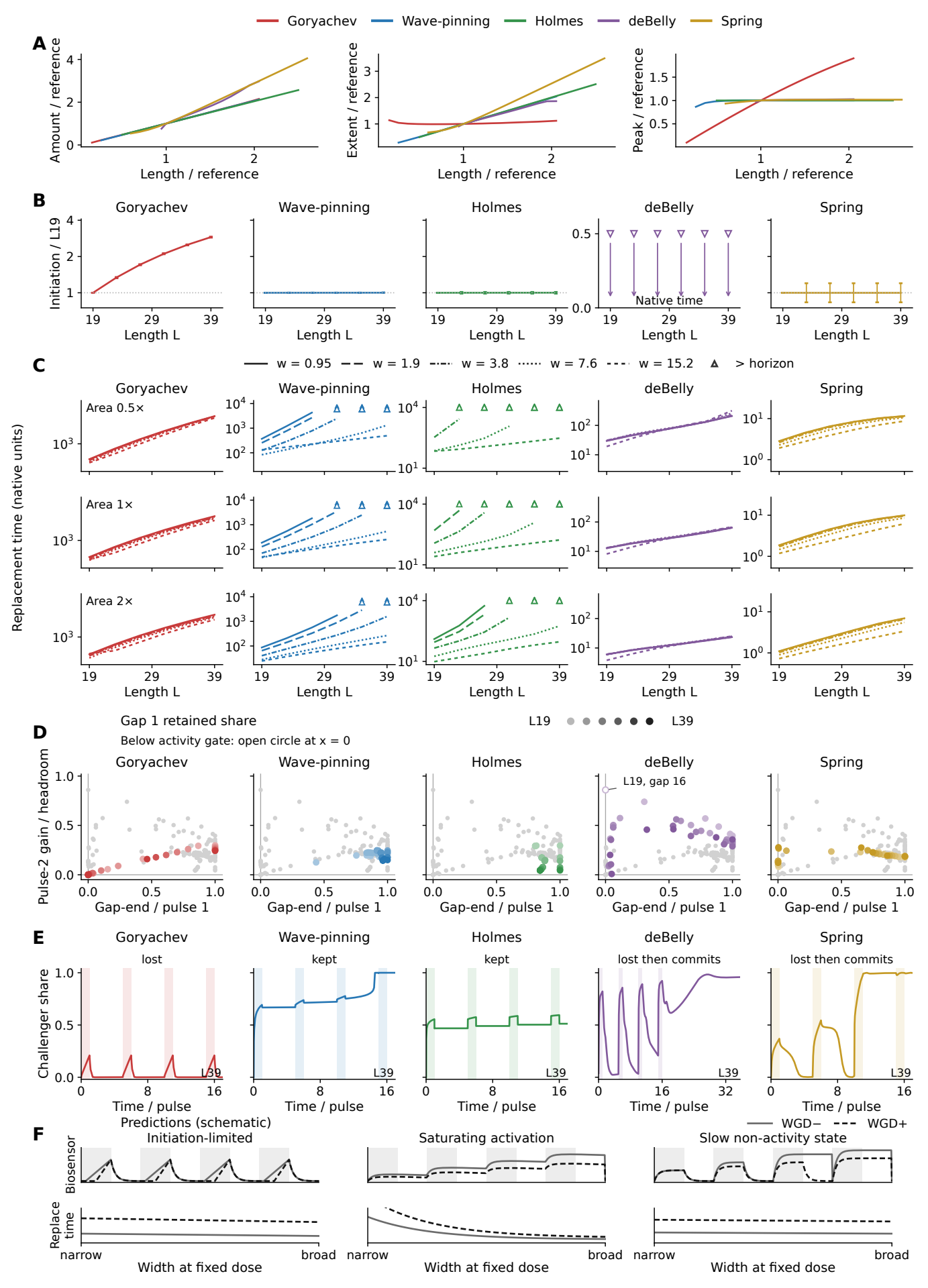
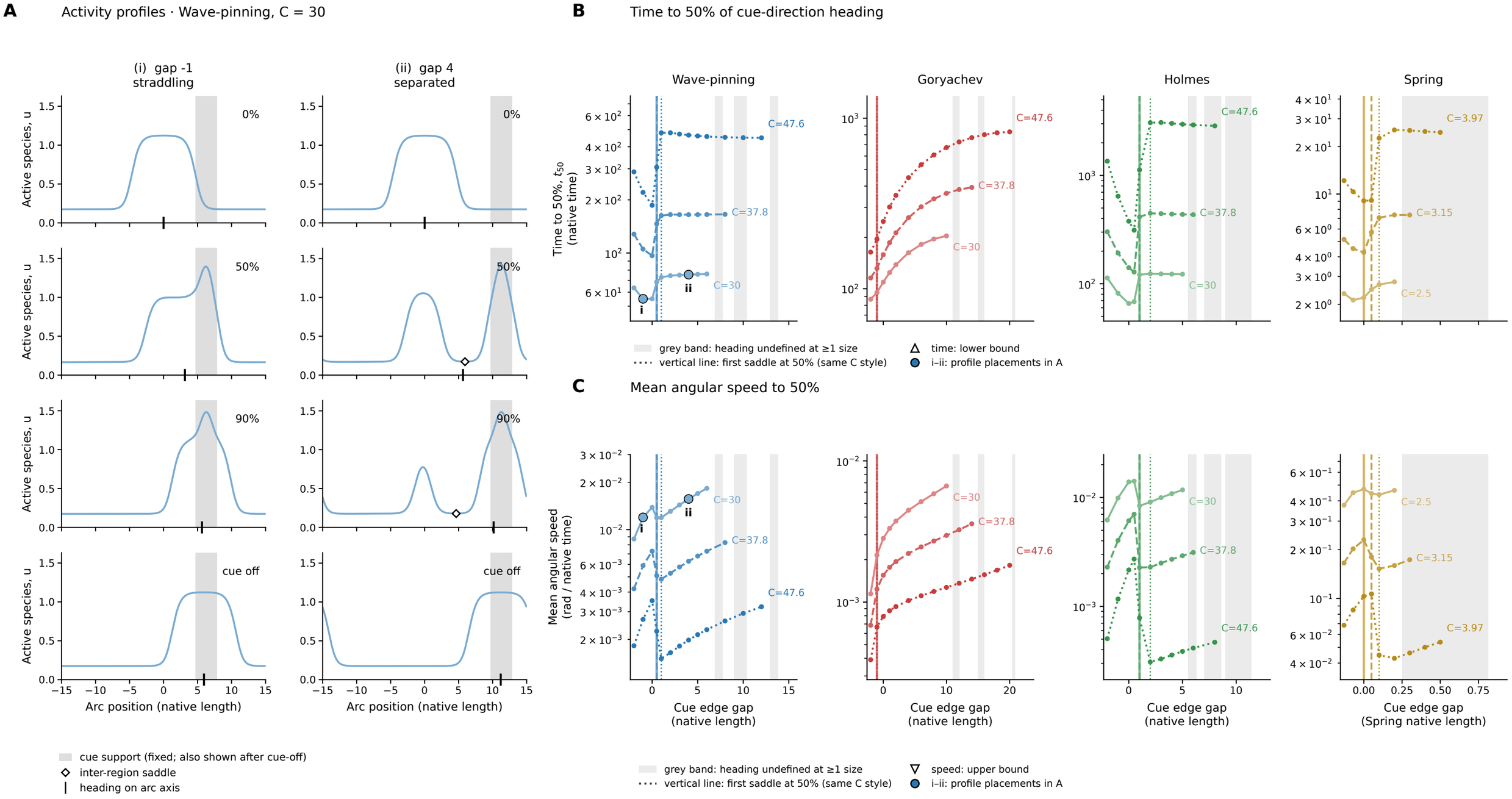
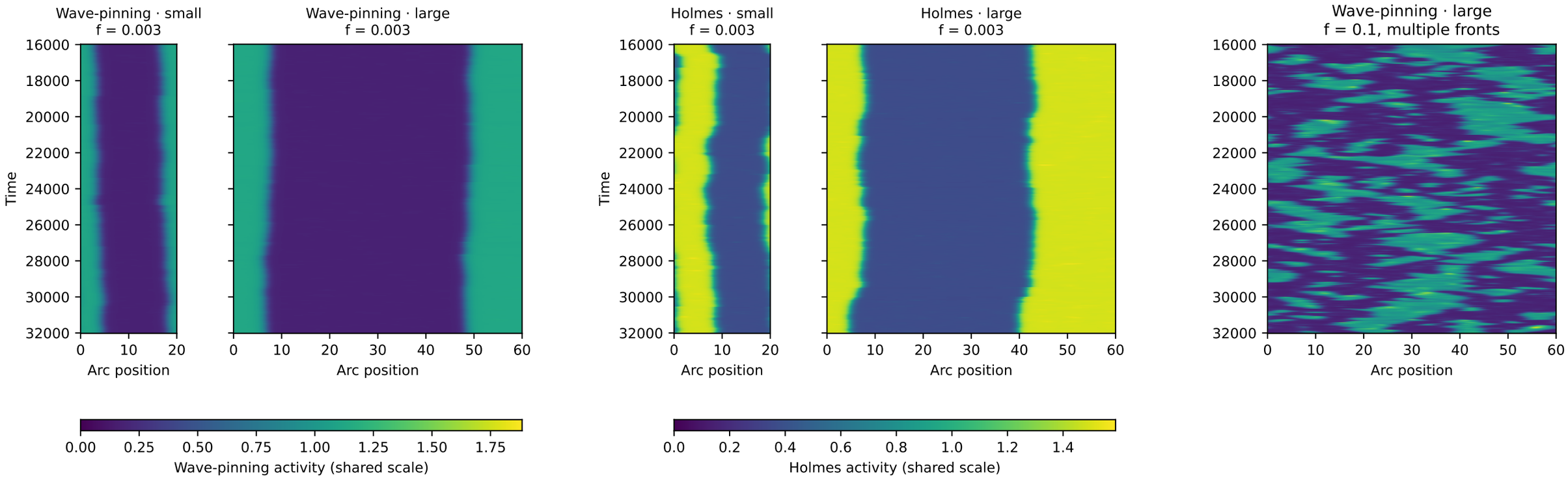

Goryachev
Wave-pinning
Holmes
deBelly
Spring
A
Amount / reference
Length / reference
Extent / reference
Peak / reference
B
Initiation / L19
Native time
Length L
C
w = 0.95
w = 1.9
w = 3.8
w = 7.6
w = 15.2
> horizon
Area 0.5×
Area 1×
Area 2×
Replacement time (native units)
D
Gap 1 retained share
Below activity gate: open circle at x = 0
L19
L39
L19, gap 16
Pulse-2 gain / headroom
Gap-end / pulse 1
E
lost
kept
lost then commits
Challenger share
Time / pulse
F
Predictions (schematic)
WGD−
WGD+
Initiation-limited
Saturating activation
Slow non-activity state
Biosensor
Replace time
narrow
broad
Width at fixed dose

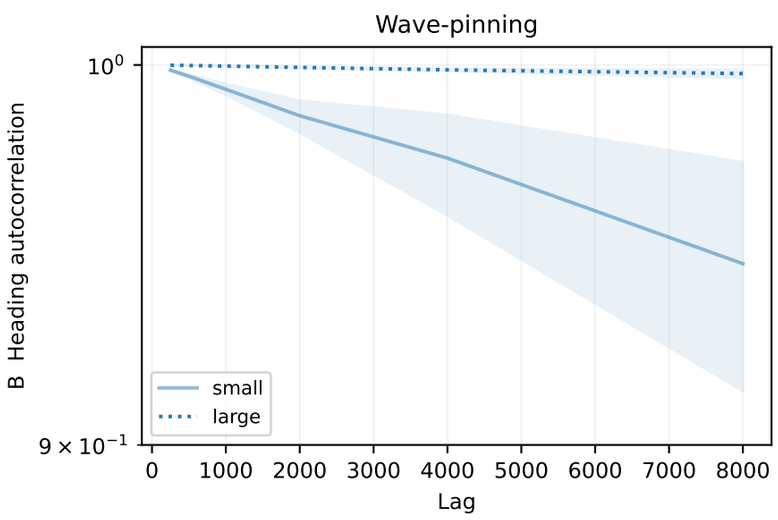

A
Activity profiles · Wave-pinning, C = 30
(i) gap -1
straddling
(ii) gap 4
separated
0%
50%
90%
cue off
Active species, u
Arc position (native length)
cue support (fixed; also shown after cue-off)
inter-region saddle
heading on arc axis
B
Time to 50% of cue-direction heading
Wave-pinning
Goryachev
Holmes
Spring
Time to 50%, t_{50}
(native time)
C=47.6
C=37.8
C=30
C=3.97
C=3.15
C=2.5
i
ii
grey band: heading undefined at ≥1 size
vertical line: first saddle at 50% (same C style)
time: lower bound
i–ii: profile placements in A
C
Mean angular speed to 50%
Mean angular speed
(rad / native time)
Cue edge gap
(native length)
Cue edge gap
(Spring native length)
speed: upper bound

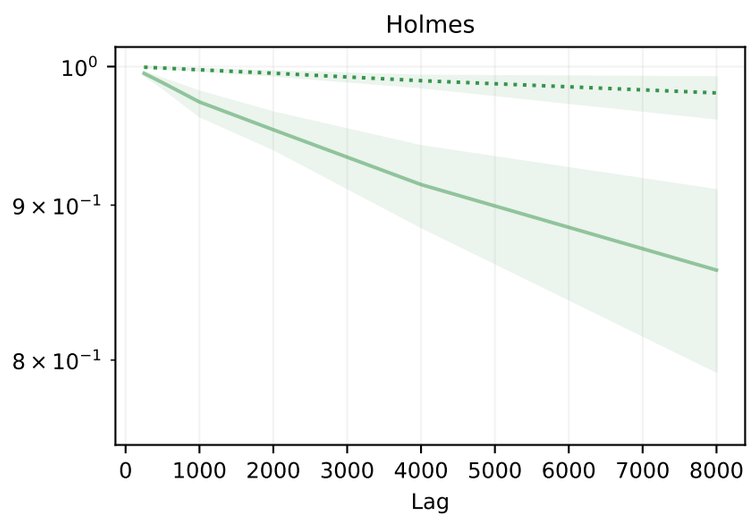
Enlarged cells hold heading longer under fluctuating input

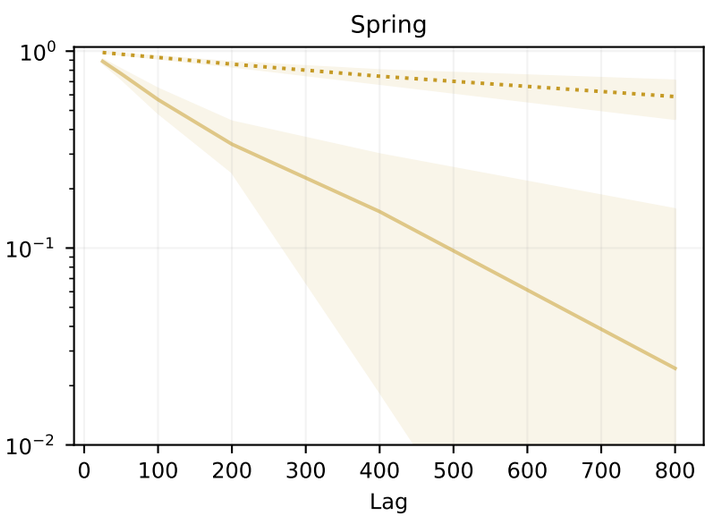
Gains reach or exceed geometry at each model's weakest measurable input; the advantage declines as input strengthens

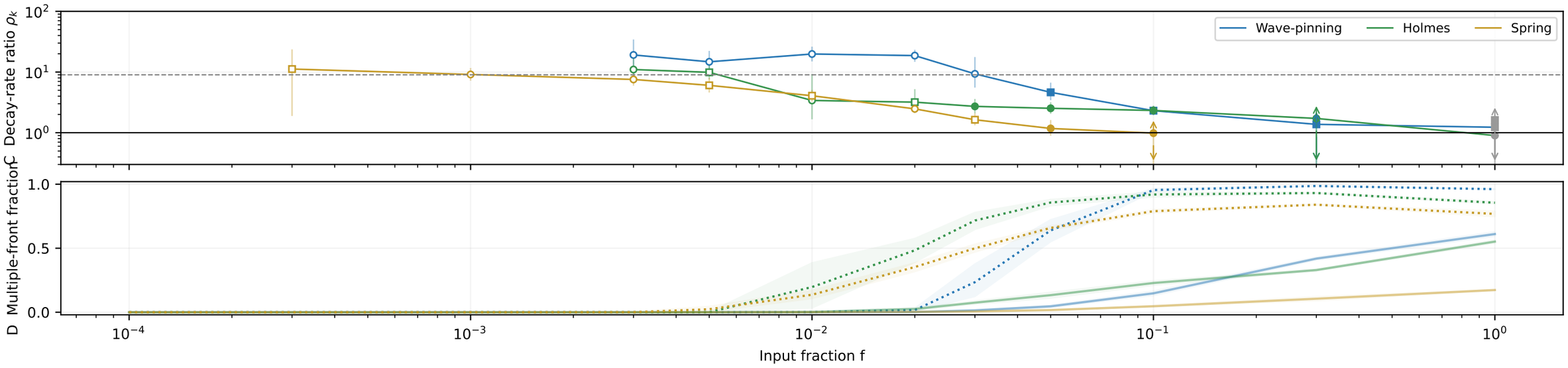
A Native activity histories

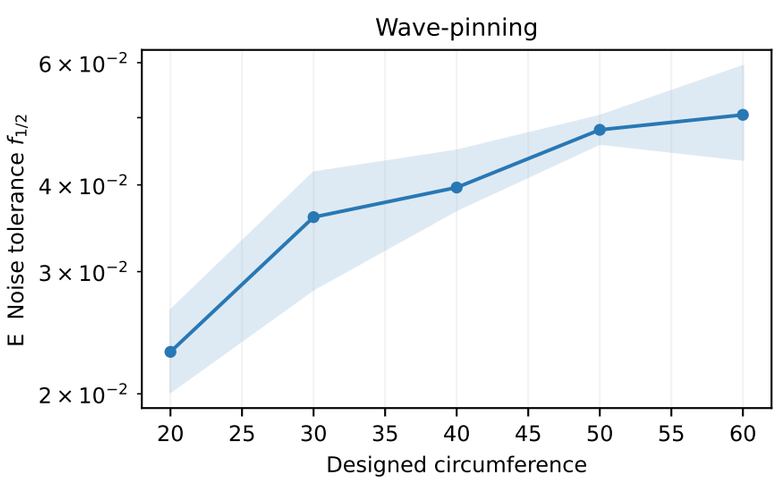


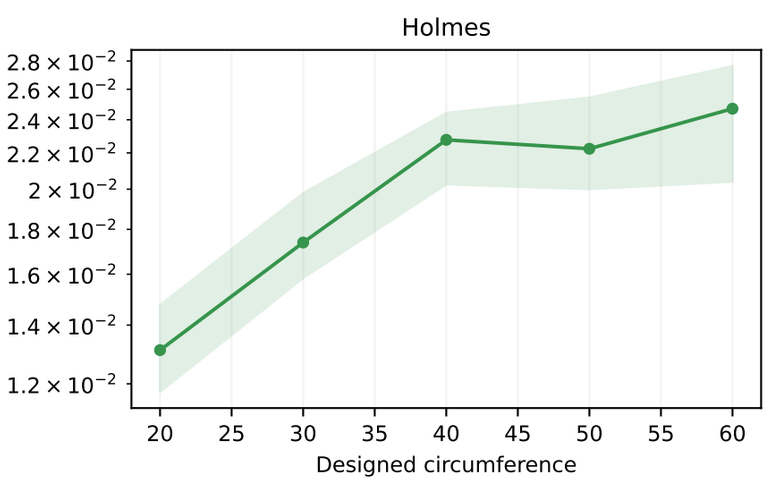


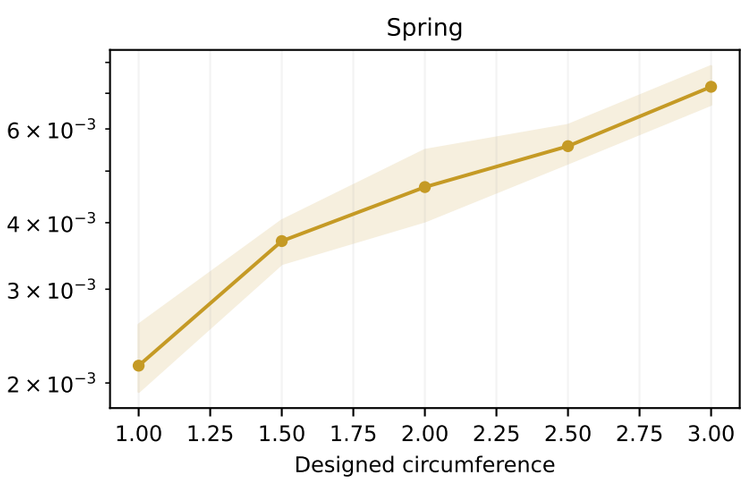












Small: lighter solid; large: darker dotted. C: open fill single front; filled multiple fronts; grey fill low contrast; square >1.5× lag drift.
Pointwise whole-seed 95% CIs. Horizontal references: ratio 1 and nominal geometry 9. D is descriptive; it does not establish a cause of erosion.
H1 uses a post hoc f < large-cell f½ domain; frozen H1 remains inconclusive. Goryachev retained in supplement only.